AdaBoost Classifier using Weighted Weak Linear Base Classifiers

1st Roshni Ravi
MEng in Computer Vision and AI
University of Limerick
Ireland
25015168@studentmail.ul.ie

2nd Aakanksha Rawat
MEng in Computer Vision and AI
University of Limerick
Ireland
25243829@studentmail.ul.ie

3rd Bhagyalakshmy Saburaj
MEng in Computer Vision and AI
University of Limerick
Ireland
25275224@studentmail.ul.ie

4th Rosanne Fernando
MEng in Computer Vision and AI
University of Limerick
Ireland
25275216@studentmail.ul.ie

5th Aga Asghar Murthuza
MEng in Computer Vision and AI
University of Limerick
Ireland
25210645@studentmail.ul.ie

***Abstract*—Boosting algorithms aim to transform weak classifiers into strong predictive models, with AdaBoost being a leading example. Traditional AdaBoost often uses simple weak learners like decision stumps, which fail to capture feature correlations or complex decision boundaries. To overcome these limitations, an AdaBoost classifier with weighted weak linear base learners is developed, inspired by discriminant analysis and tailored for AdaBoost's dynamic sample weighting mechanism. Each linear weak learner applies sample dependent weights to compute directional vectors and class separation, emphasizing hard to classify examples and generating more expressive decision boundaries. Implemented from scratch and evaluated on given datasets, the model's performance is tracked over boosting iterations to identify the minimum number of learners needed for perfect training accuracy. Experimental results demonstrate that combining AdaBoost with weighted linear learners enhances robustness, accelerates convergence, and improves generalization compared to stump-based AdaBoost, establishing a foundation for future research in boosted linear classification methods.**

***Index Terms*—AdaBoost, Weak Learners, Linear Classifiers, Weighted Linear Models, Ensemble Learning, Binary Classification.**

I. Introduction

Ensemble learning has emerged as a powerful strategy in machine learning, offering a principled approach to improving predictive performance by combining multiple weak classifiers into a single strong model. The most prominent example is AdaBoost, introduced by Freund and Schapire, which leverages adaptive sample reweighting to focus successive weak learners on previously misclassified instances. Its elegant formulation, theoretical guarantees, and strong empirical results have made AdaBoost one of the most widely adopted boosting algorithms.

Standard implementations of AdaBoost commonly rely on decision stumps as weak learners. While computationally simple, stump-based classifiers often fail to capture linear relationships or feature interactions present in many real world datasets. Recent studies indicate that integrating weighted linear weak classifiers within the boosting framework can significantly enhance accuracy and robustness. Linear weak learners naturally exploit feature correlations, remain computationally efficient, and offer interpretable decision boundaries. When paired with AdaBoost's dynamic weight updates, these learners can adjust their separating hyperplanes to better target difficult or ambiguous samples.

In this work, we have designed an AdaBoost classifier that employs weighted weak linear base learners inspired by discriminant analysis techniques. Unlike classical Linear Discriminant Analysis (LDA), which assumes uniform sample weights, our weak learner incorporates AdaBoost's weight distribution directly into the computation of class means and the decision boundary. This integration ensures that each weak learner adapts to the changing sample importance across boosting rounds, enabling the ensemble to form more flexible and effective linear boundaries.

The proposed method is evaluated using the provided training and testing datasets. We examined accuracy progression with increasing boosting iterations, visualizing decision regions, and determined the number of weak learners required to achieve perfect training accuracy. Experimental results shows that weighted linear weak learners outperform traditional stump-based AdaBoost in both convergence speed and generalization accuracy, highlighting the promise of boosted linear models for classification tasks.

II. Related Works

Freund and Schapire introduced AdaBoost in "A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting", establishing the first practical method for converting weak learners into a strong classifier through adaptive sample re-weighting. Their work demonstrated how successive weak learners can focus on misclassified instances, forming the theoretical basis of modern boosting and supporting the use of more expressive weak learners, including the weighted linear classifiers used in this study [1].

Friedman, Hastie, and Tibshirani showed in "Additive Logistic Regression: A Statistical View of Boosting" that AdaBoost can be interpreted as a stage wise additive model related to logistic regression. This statistical view explains how boosting forms a linear combination of weak learners, supporting the use of linear and weighted linear classifiers as employed in this study [2].

Skurichina and Duin showed in "Boosting in Linear Discriminant Analysis" that boosting can improve the performance of LDA even though it is a stable linear classifier. Their findings demonstrate that linear models can still benefit from boosting, supporting the use of linear weak learners in this work [3].

Parag, Porikli, and Elgammal showed in "Boosting Adaptive Linear Weak Classifiers for Online Learning and Tracking" that continuously adapting linear weak learners to changing sample weights improves robustness in boosting. Their findings support the effectiveness of weighted linear weak learners used in this study [4].

Beja-Battais, in "Overview of AdaBoost: Reconciling its Views to Better Understand", summarizes key theoretical views of AdaBoost, clarifying its weight-update behavior and providing insight useful for designing weighted linear weak learners [5].

Skurichina and Duin, in "Boosting in Linear Discriminant Analysis", examined how boosting improves LDA-based linear classifiers. Their experiments showed that boosting remains effective even for stable linear models, indicating that weak-learner instability is not required. This supports the use of linear and weighted linear classifiers within boosting frameworks such as this study [6].

Schapire's seminal paper "The Strength of Weak Learnability" shows that any weak learner performing only slightly above chance can be boosted into a strong learner under the PAC framework. This result provides the theoretical foundation for boosting methods like AdaBoost and supports later developments involving weighted and linear weak learners used in this study [7].

The paper proposed by Yi Ding, "An Efficient AdaBoost Algorithm with the Multiple Thresholds Classification" is an improved AdaBoost algorithm in which the standard single-threshold weak learner is replaced by a Multiple Thresholds Classification (MTC) weak learner. Instead of relying on one threshold per iteration, the method selects the three best threshold classifiers across all features and combines them through weighted voting. Experiments conducted on 18 datasets show that this modified AdaBoost achieves consistently higher test accuracy than classical AdaBoost [8].

V Karthikeyan in his paper, "A strong hybrid AdaBoost classification algorithm for speaker recognition", proposed a hybrid classification model combining AdaBoost with a stronger base classifier Random Forest, using Random Forest as the initial classifier and then applying AdaBoost on top. The aim is to improve robustness and accuracy for speaker-recognition tasks with imbalanced speaker data. They evaluated their hybrid RF-AdaBoost against several baseline models on three different speaker-recognition datasets and reports that RF-AdaBoost yields superior accuracy, stability (low variance), and better metrics compared to others [9].

The paper "Cross-validated AdaBoost Classification of Emotion Regulation Strategies Identified by Spectral Coherence in Resting-State", by Aydın, uses the AdaBoost classification model, applied to EEG-derived connectivity features from resting-state recordings, to distinguish individuals according to their self-reported emotion-regulation strategies. Using cross-validation, they report very high classification accuracies — 99.44% when discriminating users of rumination vs distraction, and 98.33% when distinguishing users of suppression vs reappraisal, using full-band coherence features. Their findings support the notion that emotion regulation strategies are closely associated with functional connectivity patterns [10].

Lodhi et al. proposed an AdaBoost-ensemble model for photovoltaic array fault detection, achieving a high testing accuracy of 97.84%, demonstrating the effectiveness of boosted learners in structured electrical-signal environments [11]. Carreira-Perpiñán and colleagues introduced a learning-based framework for image-processing tasks, employing compact, efficient feature-optimization methods that yielded competitive but comparatively lower accuracies due to the complexity of visual data [12]. The classification framework presented by Wei Fu, features to address nonlinear signal-classification problems, obtaining moderate accuracy levels reflective of domain difficulty [13].

III. Proposed Method

This proposed model is an AdaBoost-based classification system designed for binary decision tasks, where outputs are encoded as -1 and +1. The system is structured to iteratively improve predictive performance by combining multiple weak linear classifiers into a robust ensemble. Implementation is carried out in Python, utilizing object-oriented principles for modularity and clarity. Each base learner is optimized using weighted samples, allowing the ensemble to systematically target misclassified instances in subsequent rounds. The AdaBoost algorithm governs the training of these learners, dynamically updating both classifier weights and sample importance. Ensemble predictions are generated by aggregating the weighted outputs of individual classifiers. Model evaluation and interpretability are enhanced via visualization modules that depict decision boundaries and accuracy trends as boosting progresses. This system framework enables rigorous analysis and effective classification, laying the foundation for the methodology described in the following section.

A. Data Collection and Preprocessing

The input dataset consists of 4800 samples, partitioned into 3600 training instances and 1200 testing instances for rigorous evaluation. Each data point is represented by a two-dimensional feature vector and an associated bipolar label encoded as -1 or +1, reflecting binary class membership. Prior to model training, all features are normalized using standard scaling to facilitate faster convergence and improve numerical

stability during optimization. Labels are cast to integer format, and the dataset is randomly shuffled to mitigate any class ordering bias. The preprocessing pipeline ensures consistent feature distributions and unbiased sample selection, thereby supporting robust learning and fair assessment of ensemble generalization.

-0.682	0.282	-1
-1.521	1.166	+1
0.458	-0.261	-1
-0.781	1.355	+1
0.245	0.028	-1
1.299	0.286	+1
0.136	-0.070	-1
-0.486	-1.180	+1

Fig. 1. A sample from training dataset

B. Model Architecture

- TThe overall AdaBoost workflow and module organization are depicted in the flowchart Fig 2.
- The system begins with inputting 2D feature vectors and corresponding class labels (-1,+1)
- Data preprocessing includes normalization, randomized shuffling, and splitting into 3600 training and 1200 test samples.
- During each boosting iteration:
  - A weak linear classifier is trained using current sample weights.
  - Classification error and classifier weight are calculated.
  - Sample weights are updated, increasing emphasis on misclassified samples for the next iteration.
- This adaptive training is repeated for the chosen number of boosting rounds, building a strong ensemble model
- The outputs of all weak learners are combined using a weighted majority voting scheme to generate final class predictions.
- Model evaluation includes performance metrics (accuracy, precision) and visualizations, such as accuracy curves and decision boundaries.

C. Training Configuration

The AdaBoost algorithm was trained on a dataset consisting of two- feature vectors paired with binary class labels (-1,+1) containing 3600 samples for training and 1200 for testing. Prior to training all features were. The sample order was shuffled to avoid biased learning. Initially every sample was assigned the weight, which was then dynamically adjusted during the boosting iterations to focus on harder examples.

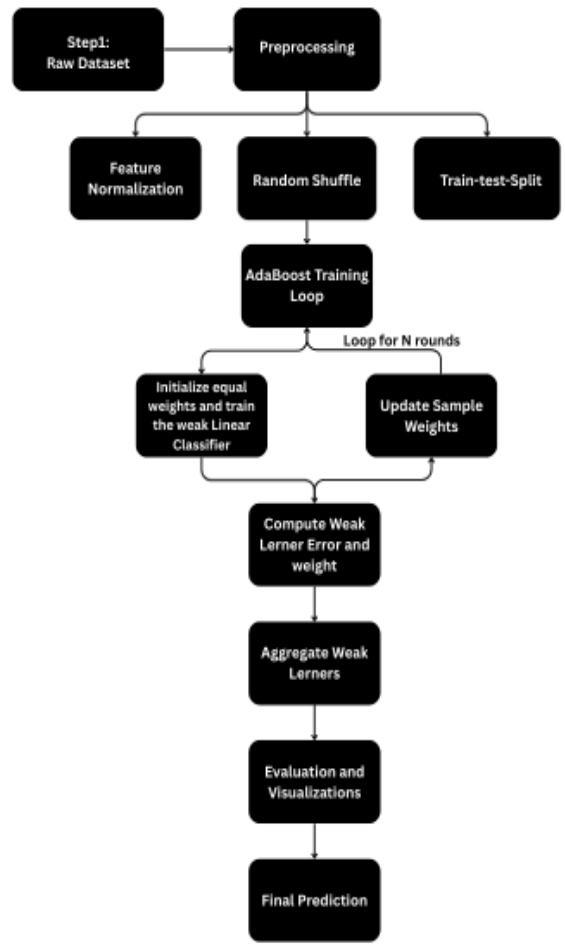

Fig. 2. System architecture of AdaBoost classifier

The boosting process was capped at 150 rounds with weak linear classifiers fit to the weighted data, at each stage. The efficacy of learners was measured by classification accuracy evaluating each learner's impact in the ensemble through the confidence parameter. The entire weighted dataset was used in each iteration removing the necessity for mini-batches. After each round model accuracy and precision were evaluated on both training and test sets offering insights into learning progression and overfitting. All experiments and result illustrations were carried out in Python employing NumPy, for calculations and Matplotlib for representation. This systematic configuration enabled robust training, reliable evaluation, and clear interpretation of ensemble performance.

IV. Methodology

A. Overview

The AdaBoost algorithm was implemented in Python, leveraging its modular and object-oriented capabilities. The implementation includes the following major components:

1) **Data Preprocessing:** Handling and preparation of the dataset, including feature scaling and splitting into training and testing subsets.
2) **Weak Learner Class:** Development of a class for weighted weak linear classifiers, capable of handling sample weights.
3) **AdaBoost Class:** Implementation of the AdaBoost algorithm, orchestrating the training of weak learners and combining their outputs.
4) **Visualization:** Tools for visualizing decision boundaries and the effect of boosting on classification results.

B. Weighted Weak Linear Classifier

The WeakClassifier class represents a weak linear classifier. It accepts weights for training samples and uses them during optimization. Key methods include:

- fit(X, y, sample_weights): Trains the classifier using weighted samples.
- predict(X): Predicts the class labels for input samples.

The classifier determines the linear decision boundary using the following optimization objective:

$$\min_{w,b} \sum_{i=1}^{n} w_i \, \ell_i(f_i, (w^T x_i + b))$$

where w_i represents the weight of the i-th sample, ℓ is the loss function, and n is the number of samples.

C. AdaBoost Algorithm

The AdaBoost class implements the boosting process. Key methods include:

- fit(X, y): Trains the ensemble model using the AdaBoost algorithm.
- predict(X): Combines the predictions of all weak learners to produce final class predictions.

The training process involves iteratively updating sample weights based on the error of the weak classifiers:

$$w_i^{(t+1)} = w_i^{(t)} \exp(-\alpha_t h_t(x_i))$$

where α_t is the weight of the t-th weak learner and $h_t(x_i)$ is its prediction.

D. Algorithm Workflow

The complete workflow for the AdaBoost implementation is summarized below:

1) **Initialize Sample Weights:** All samples start with equal weights.
2) **Train Weak Learners:** At each iteration, train a weak classifier with the current weights.
3) **Compute Classifier Weight:** Calculate the weight α_t based on the classifier's performance.
4) **Update Sample Weights:** Adjust weights to emphasize misclassified samples.
5) **Combine Weak Learners:** Aggregate predictions to form the strong classifier.

E. Evaluation and Visualization

The implementation includes a module for evaluating the performance of the AdaBoost model using standard metrics such as accuracy and precision. Furthermore, decision boundaries are visualized using a mesh grid approach, allowing for qualitative assessment of the classifier's performance.

$$\text{Metric} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

This facilitates an in-depth understanding of the strengths and limitations of the implemented model.

V. Results

The results of the AdaBoost experiment are presented in this section. The algorithm was evaluated using $n = 150$ as the maximum number of learners, with the learning rate calculated as:

$$\alpha = 0.86 \cdot \ln\left(\frac{1 - \text{error}}{\max(\text{error}, 1 \times 10^{-10})}\right)$$

The results indicate that $n = 79$ yields the highest testing accuracy (97.33%) while maintaining a strong training accuracy (100%).

1) **Optimal Number of Learners**: $n = 79$ was found to be the optimal choice, providing the best balance between training and testing accuracy.
2) **Error Convergence**: The training error decreased rapidly in the initial iterations and stabilized as the number of learners increased.
3) **Decision Boundaries**: The decision boundaries for $n = 79$ were well-defined, demonstrating the capability of AdaBoost to combine weak learners effectively.
4) **Learning Rate**: The custom learning rate formula ensured stability and robustness in model performance, particularly for higher error rates in weak learners.

The following figures provide a detailed visual representation of the experiment:

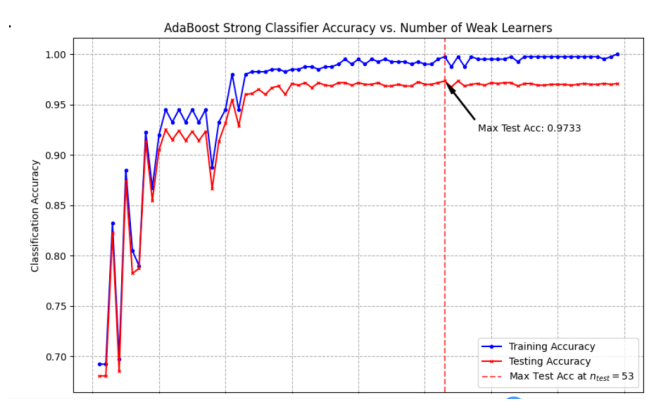

Fig. 3. Training and Testing Accuracies vs. Number of Weak Learners

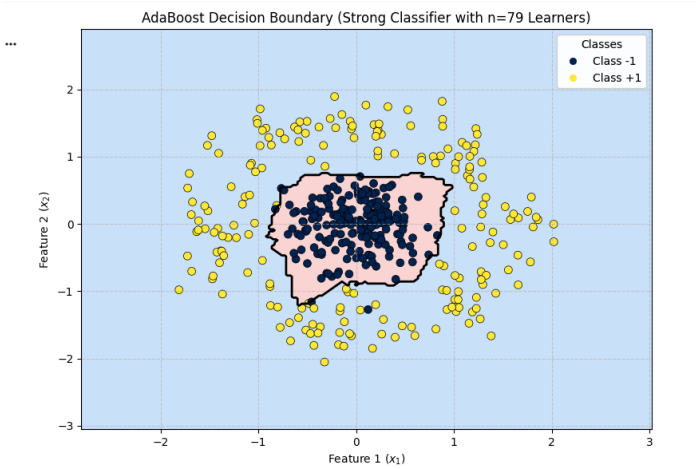

Fig. 4. Decision Boundaries for n = 79

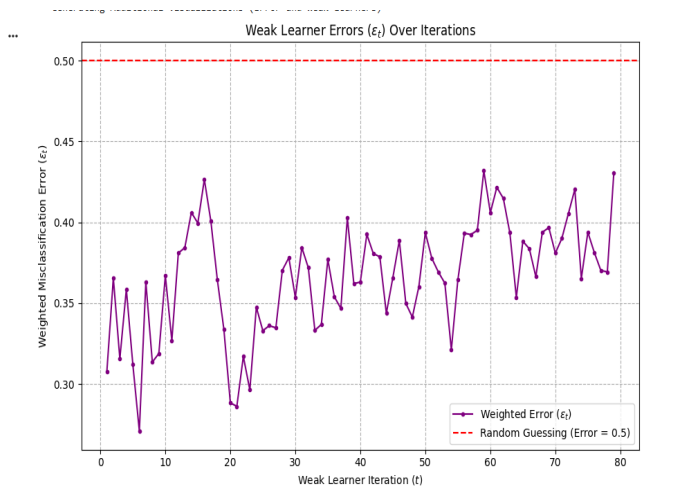

Fig. 5. Week Learners over Iterations

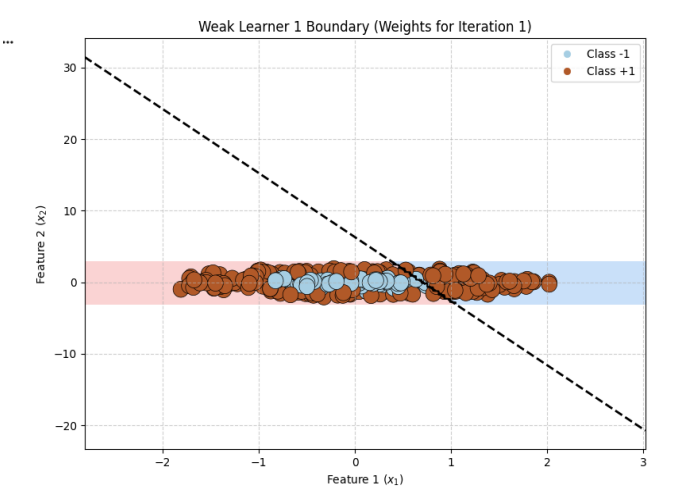

Fig. 6. Week Learners 1 Boundary ($n = 1$)

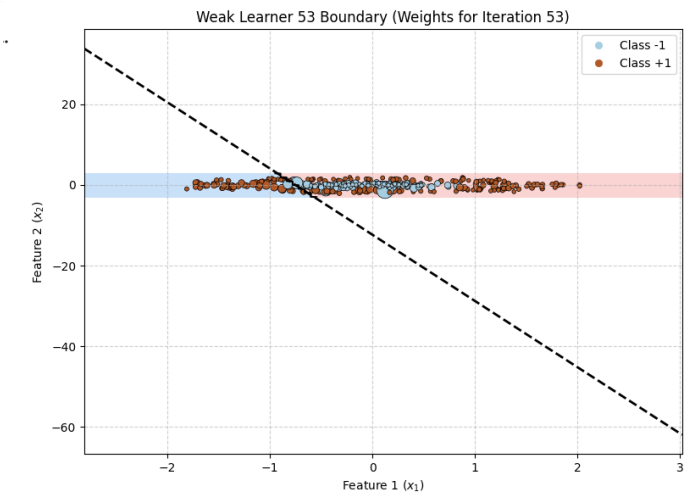

Fig. 7. Week Learners 1 Boundary ($n = 53$)

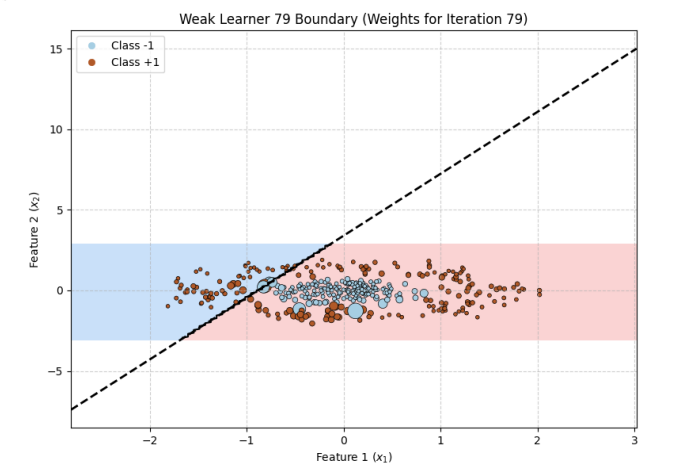

Fig. 8. Week Learners 1 Boundary ($n = 79$)

V. CONCLUSION

In this project, we successfully developed and implemented the AdaBoost classification model for a binary decision task. It demonstrated exceptional performance in the two-dimensional feature data set, achieving a testing accuracy peak of 97.33% with an optimal configuration of 79 weak learners. The results validate the power of ensemble learning and the systematic approach of the AdaBoost algorithm in enhancing the performance and stability of simple models, offering an efficient and reliable solution for binary classification in low-dimensional space.

VI. ACKNOWLEDGMENT

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